



Big Companies Also Get To **LEARN** From **STARTUPS** About The Actual Needs Of The Market



Hwai Lin Khor

head of sales and business development for digital security solutions, Infineon Asia-Pacific



Adoption of Industry 4.0 has been slow in India. But advancements in electronics, the Internet of Things (IoT) and correlated technologies indicate a soon-to-be mainstream automated factory ecosystem. Hwai Lin Khor, head of sales and business development for digital security solutions, Infineon Asia-Pacific, talks to Baishakhi Dutta about technical upgrades and corresponding design improvements that the industry is following to embrace Industry 4.0

Q. How can design engineers come up with more innovative solutions?

A. First, they should not take too long to make their solution perfect.

Second, it is good to work with big companies that encourage startups to work with them and gain industry knowledge, and maybe get customers who need their solutions.

Big companies also get to learn from startups about the actual needs of the market. This also helps in product development.

Third, when working on the IoT, they need to design security from the beginning to avoid re-design.

Q. Do you think the R&D system is well-developed in India?

A. People working in MNCs that have established their R&D centres in India are trained and know what to do. Once local engineers also set up their own companies and offer solutions, there will be a lot more decision-making taking place in India, where local players will have the power to innovate.

Q. Is the IoT affecting the ESDM sector?

A. The IoT is a very broad term. In fact, many applications are part of the IoT and more will join soon. There will be a lot of exploration, and different applications will have different requirements.

Moreover, interoperability is a challenge as every company has its own proprietary standards at this moment.

Q. How is India coping with Industry 4.0?

A. In India it is already possible to have a dark factory, because currently,

one does not need to go through all the previous stages. Some innovative players who have started factories with Industry 4.0 capability need expertise from other foreign players who have implemented it.

Q. Where do you see the Indian market while talking about demand for your products?

A. We have seen Make in India move due to which there are a lot of factories coming up. This includes not just manufacturing but also R&D for local markets. R&D taking place in the Indian market is good for Infineon because we can work together with companies, look into product designs, sell components to them and develop solutions for the Indian market and neighbouring countries like Bangladesh and Sri Lanka, who import Indian technologies.

Also, with many startups coming up and various innovative business models being tried out, there are a lot of opportunities here.

Q. Any plans of tying up with academia?

A. We have a university relation group, including in India, which gets in touch with professors and schools to look into their curriculum, and decides how we can work with them.

Together with university faculty, Infineon co-organises competitions like hackathons and topic-specific workshops with students.

Q. How are you coming up with effective designs in terms of evolving designs, security and services?

A. Infineon has been in the security domain for more than thirty years. We started with the smartcard industry

where we sold our security controller to various applications, such as government, payment and transportation. Now, we are seeing many IoT applications that require high security and are also looking to secure their network and devices to avoid attacks.

We have seen strong demand for security for IoT applications and, hence, used our smartcard industry knowledge and applied it to embedded systems so that it is easy for engineers to build secure IoT devices.

Q. How are players like you improving software and hardware integration to optimise embedded systems?

A. By offering solutions that combine hardware with software to enable easy integration and ease of use. The security solutions are designed to reduce integration effort and support faster time-to-market.

Q. Which industrial trends will rule the market in the surveillance sector? How will Infineon's products impact it?

A. In Industry 4.0, a lot of automation is happening and, so fewer human and more machines are offering auto self-regulating systems. For such systems, you need an intelligent controller to help process the information. At the same time, information is being transferred from point to point. So, you need security to be top-notch.

I see Infineon contributing in many areas, from powering a factory with energy-efficient solutions, motor drives that drive the equipment and robots, sensors that allow machines to be intelligent and secure to enable devices to communicate securely with one another. **EFY**